CPT205 Assessment 1

A two-dimensional (2D) Greeting Card for XJTLU 20th Anniversary

Module	CPT205 Computer Graphics
Degree Programme	Information and Computing Science
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1. Brief Description

The title of this greeting card is **"XJTLU 20th Anniversary Greeting Card"**. It consists of two main stages: an **envelope cover scene** and an **interactive celebration scene**.

The cover scene shows a pink envelope completely covering the window. When user clicks, the envelope will open and then entering the main celebration view.

Figure 1: The Cover Scene

The interactive scene includes XJTLU's Central Building (CB), a tree, flowers, some drifting clouds, balloons, and a rotating sun. User can click the floating box labeled "Click me!" right above the tree to reveal a rising birthday cake celebrating the 20th Anniversary of XJTLU.

Figure 2: The Interactive Page

2. Features

2.1 Window and Projection Setup

The program initializes a 1000×600 window and sets a 2D orthogonal projection via `reshape()` using `gluOrtho2D()`. Alpha blending (`glEnable(GL_BLEND)`) provides smooth transparency effects.

2.2 Envelope Cover Scene

The opening envelope is drawn in `drawIntro()` using layered `GL_TRIANGLES` and `GL_QUADS`. The top triangle flap uses a darker pink tone to mimic folded paper, while lighter tones are applied to the lower parts. Text "*To XJTLU*" is drawn via `drawText()` using the `GLUT_BITMAP_TIMES_ROMAN_24` font. A short message "*Click to open the greeting card*" guides user interaction below the envelope. When user clicked, the variable `introOpening` triggers an expansion animation controlled by `timerFunc()`.

2.3 Background and Layout

In the (`drawScene()`) function, I used `GL_QUADS` to draw a soft gradient background from light pink to sky blue, while a subtle pink frame is added with `GL_LINE_LOOP` to simulate a real card border.

2.4 Cloud and Sun Animation

Clouds3 are initialized in `initClouds()` and generated in `drawCloud()` using multiple circular fans that overlap for a fluffy look. Their motion combines horizontal drifting and sine-based vertical offset, adjusted by keyboard input (`A`, `D`, and `S`). The sun4, drawn in `drawSun()`, is a circle with eight line rays. Its rotation is rotated by pressing key `K` and animated through the variable `sunRotation`.

2.5 Tree and Interactive Box

The tree5 is drawn in `drawTree()` using brown rectangular trunks and layered green circular fans as foliage. Above it floats a clickable box generated inside `drawScene()` that gently oscillates using a sine function. When clicked the box, it sets `cake.visible = true`, triggering the cake animation in the next frame.

Figure 3: Clouds

Figure 4: Sun

Figure 5: Tree and Floating Box

2.6 Central Building (CB)

The XJTLU Central Building6 is drawn in `drawCB()` using a combination of `GL_LINE_LOOP` and `GL_LINE_STRIP` for its outlines. A soft glow effect is achieved using a sine-based brightness oscillation. The phrase "*Lights and Wings*" appears on top of the CB, rendered by `drawText()` to symbolize inspiration and innovation.

2.7 Flowers and Balloons

Flowers7 are drawn via `drawFlower()`, which combines stems, leaves and six circular petals arranged radially. Balloons8 are created dynamically by `spawnBalloon()` and rendered by `drawBalloon()`. Their floating motion uses a sine-based lateral sway, and colors are converted from HSV to RGB using `hsv2rgb()` function for smooth color diversity.

2.8 Cake Animation

The celebration cake9 is controlled within `drawScene()` and `timerFunc()`. It is composed of two layers. There are two candles[1] on the top and a banner at the bottom. The banner "Happy 20th Anniversary of XJTLU" is generated via `drawText()` above the cake once it rises completely.

[1]Each candle represents a period of 10 years. This symbolizes that the school has gone through two decades since its establishment and has continuously accumulated experience and achievements.

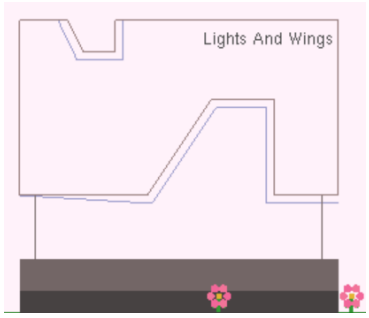


Figure 6: Central Building (CB)

Figure 7: Flowers

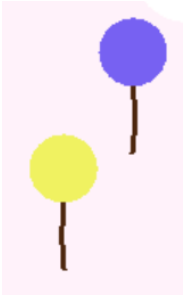


Figure 8: Balloons

Figure 9: Celebration Cake

2.9 Text Rendering and Help Window

All text rendering is handled by `drawText()`, which internally calls `glutBitmapCharacter()` to draw each character at specified 2D coordinates. At the bottom of the greeting card, I set a help section, which is mainly used to assist users in understanding how to use this greeting card.

Figure 10: Help Information Bar

3. User Instruction

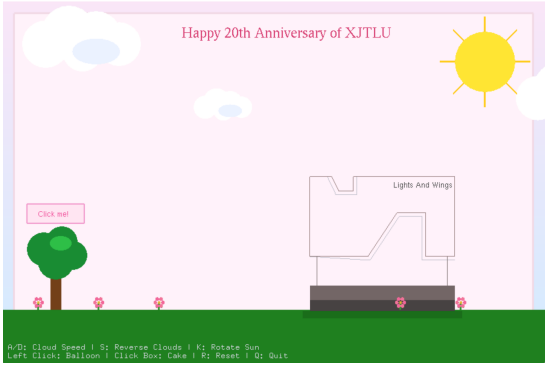
Keyboard Interaction

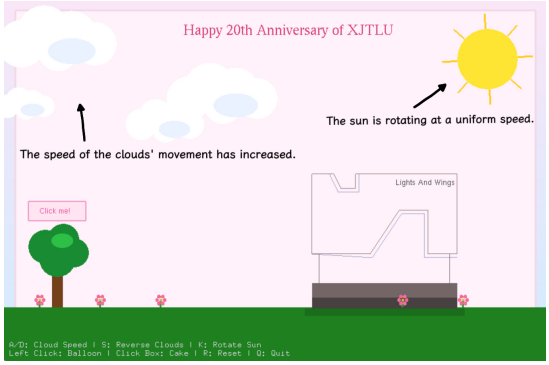
- **A(a) / D(d)**: Adjust cloud speed ('A'/'a' to slow down, 'D'/'d' to speed up).
- **S(s)**: Reverse cloud direction.
- **K(k)**: Sun rotates / stops rotating.
- **R(r)**: Reset the scene.
- **Q(q) / ESC**: Quit the program.

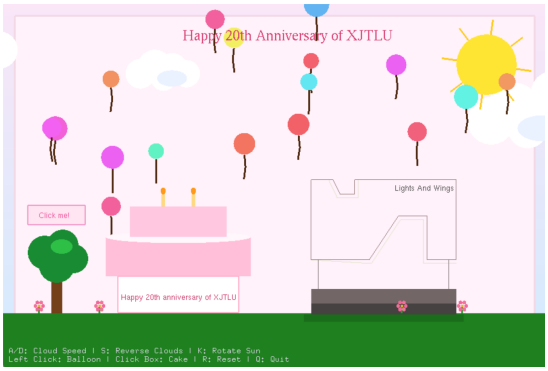
Mouse Interaction

- Click on the envelope to open the card.
- Click on the floating "Click Me!" box above the tree to see the cake.
- Click anywhere else in the scene to generate colorful balloons.

4. Typical Screenshots







5. Conclusion

This 2D greeting card is designed to celebrate the 20th anniversary of XJTLU through the combined application of geometric modeling, animation and interaction. The card not only achieves visual beauty but also includes functionality and interaction using the keyboard and mouse. However, there are still several aspects that can be improved in this project. The program's interaction logic only involves mouse and keyboard inputs, lacking a multi-level user feedback mechanism. Additionally, the performance of the animations mainly relies on geometric movement and color changes, if there is possible to incorporate some sound elements could be beneficial. Here is a direction that I can further explore and optimize in the future.